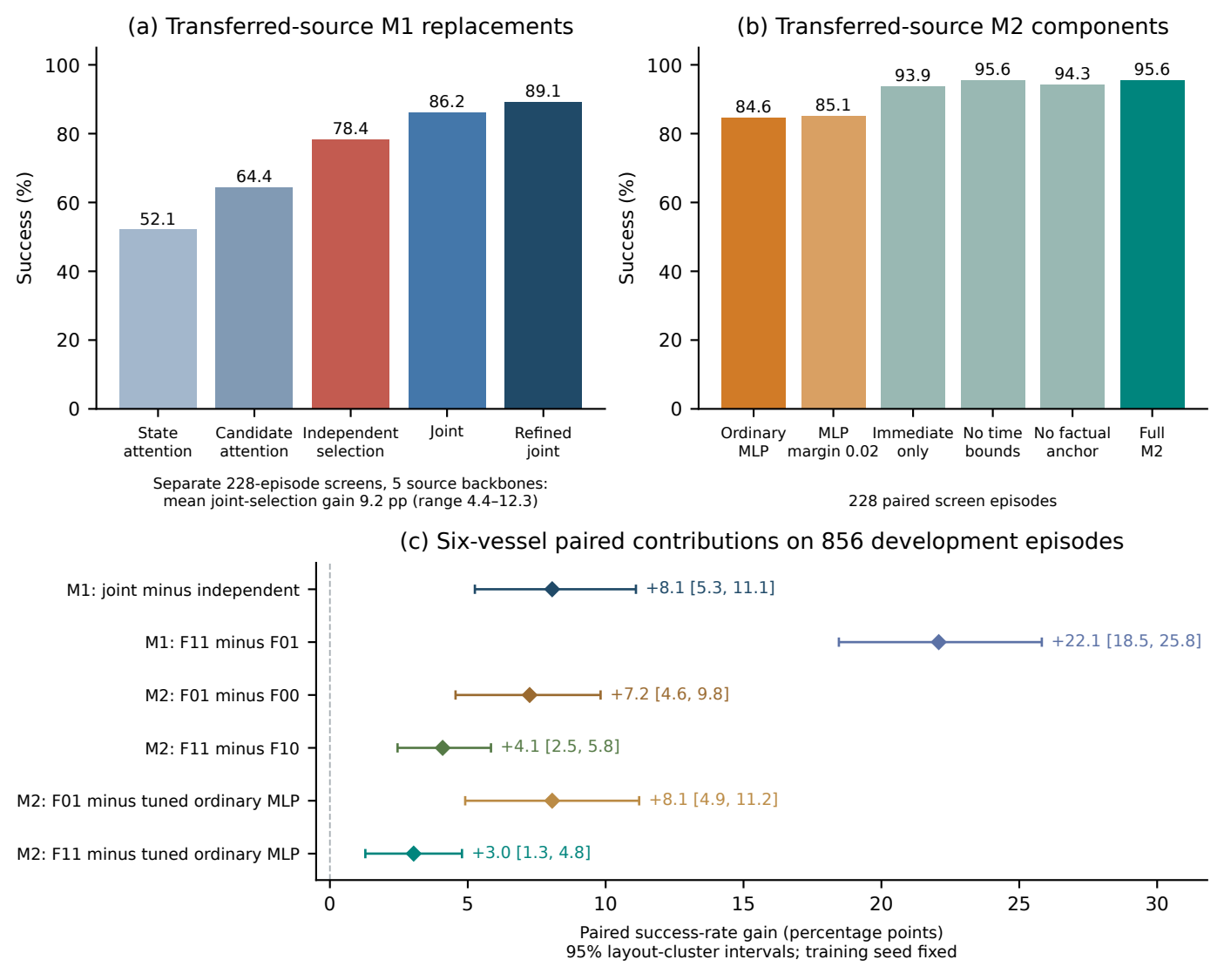

(a) Transferred-source M1 replacements
Success (%)
52.1
64.4
78.4
86.2
89.1
State attention
Candidate attention
Independent selection
Joint
Refined joint
Separate 228-episode screens, 5 source backbones:
mean joint-selection gain 9.2 pp (range 4.4–12.3)
(b) Transferred-source M2 components
84.6
85.1
93.9
95.6
94.3
95.6
Ordinary MLP
MLP margin 0.02
Immediate only
No time bounds
No factual anchor
Full M2
228 paired screen episodes
(c) Six-vessel paired contributions on 856 development episodes
M1: joint minus independent
+8.1 [5.3, 11.1]
M1: F11 minus F01
+22.1 [18.5, 25.8]
M2: F01 minus F00
+7.2 [4.6, 9.8]
M2: F11 minus F10
+4.1 [2.5, 5.8]
M2: F01 minus tuned ordinary MLP
+8.1 [4.9, 11.2]
M2: F11 minus tuned ordinary MLP
+3.0 [1.3, 4.8]
Paired success-rate gain (percentage points)
95% layout-cluster intervals; training seed fixed